

《计算机网络》作业-2025 年

第一章作业：（英文版教材第一章 3, 10, 11, 16, 17, 18, 30）

3.(1.1 QoS) Besides bandwidth and latency, what other parameter is needed to give a good characterization of the quality of service offered by a network used for (i) digitized voice traffic? (ii) video traffic? (iii) financial transaction traffic?

10(1.3.1 Protocol Hierarchies). What are two reasons for using layered protocols? What is one possible disadvantage of using layered protocols?

11(1.3.3 Connection-Oriented Versus Connectionless Service). What is the principle difference between connectionless communication and connection-oriented communication? Give one example of a protocol that uses 1) connectionless communication 2) connection-oriented communication.

16(1.4 Reference Models). Which of the OSI layers and TCP/IP layers handles each of the following:

- 1) Dividing the transmitted bit stream into frames
- 2) Determining which route through the subnet to use.

17(1.3.1 Protocol Hierarchies). If the unit exchanged at the data link level is called a frame and the unit exchanged at the network level is called a packet, do frames encapsulate packets or do packets encapsulate frames? Explain your answer.

18(1.3.1 Protocol Hierarchies). A system has an n -layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h -byte header is added. What fraction of the network bandwidth is filled with headers?

30(1.3.5 The Relationship of Services to Protocols). Suppose there is a change in the service (set of operations) provided by layer k . How does this impact services at layers $k-1$ and $k+1$?